Riccardo Buscicchio | Curriculum Vitæ

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Relativistic astrophysicist, developing advanced data analysis and statistical frameworks for Bayesian and frequentist methods. Applications include space-mission modelling, signal detection and parameter estimation in gravitational-wave astronomy, population inference, stochastic background searches.

Contacts

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Website & publications record: www.riccardobuscicchio.com – arXiv – ORCID

Academic positions

Università degli Studi di Milano-Bicocca **Milan, Italy**
Postdoctoral scholar (Assegnista di ricerca), Department of Physics "G.Occhialini" *2021 - current*

- *Main activity*: development of LISA data analysis ground-segment for the Italian Space Agency.

Education

University of Birmingham **Birmingham, UK**
Ph.D., School of Physics & Astronomy *2017-2021*

- *Supervisor*: A. Vecchio. Thesis resulted in 6 short-author publications.
- *Thesis Title*: Topics in Bayesian population inference for Gravitational Wave Astronomy

This thesis explores a number of topics related to Bayesian inference in gravitational-wave astronomy. From hierarchical inference on population of stellar mass binary black hole mergers, to the development of an end-to-end parameter estimation routine for space-based interferometers. Other topics are investigated: population of binary white dwarfs in satellite galaxies of the Milky Way; constraints from stochastic background on lensing of gravitational waves from binary neutron star and binary black hole mergers; statistical techniques for simultaneous inference on multiple undistinguishable sources.

Università degli Studi di Pisa **Pisa, Italy**
Master's degree in Theoretical physics *2013-2016*

- *Final degree grade*: 110/110
- *Supervisor*: G. Cella. Thesis resulted in one short-author publications.
- *Thesis title*: An improved detector for non-gaussian stochastic background of gravitational waves.

This thesis explored the idea of using functional formalism from stochastic processes and classical field theory to develop a new detection algorithm, with improved performance, for non-gaussian stochastic backgrounds of gravitational waves

Columbia University **New York NY, USA**
INFN-NSF Summer Internship *Jun-Sept 2013*

- *Supervisor*: S. Marka, I. Bartos.

We estimated the contribution to noise level in second and third generation ground-based detectors due to primary and secondary cosmic ray showers impinging on the interferometer mirrors.

Università degli Studi di Pisa **Pisa, Italy**
Bachelor's degree in Physics *2008-2012*

- *Final degree grade*: 109/110.
- *Thesis title*: Template banks for gravitational wave detection: an application of Information Geometry.

This thesis explored the idea of using differential geometry formalism (as defined in the context of Information theory) to develop a template placing algorithm over source parameter space with non-trivial manifold structure.

Metrics

Publications:

19 short-author papers published in major peer-reviewed journals
(out of which **5** first-authored papers).
12 collaboration papers, with substantial contribution, published in major peer-reviewed journals
5 papers in submission stage,
2 other publications (thesis, white papers, long-authorlist reviews)

Total number of citations: >10000. **h-index**: 21 (using ADS and iNSPIRE).
Web links to list services: ADS; iNSPIRE; arXiv; orcid.
Full list of publications available below and at `www.riccardobuscicchio.com/publications`.
Full list of presentations available below and at `www.riccardobuscicchio.com/talks`.

Grants, Prizes, & Awards

Career prizes:

- Braccini PhD Thesis Prize, Gravitational Wave International Committee honorable mention. *2021*
- Michael Penston PhD Thesis Prize, Royal Astronomical Society runner-up prize. *2021*

Other funding:

- EuroHPC PRACE "LISAS-FIT" proposal, 100k CPUh on Leonardo BOOSTER *2023*
- Google Cloud for Researchers, 4kEUR Google Cloud Research Credits *2023*
- CINECA ISCRA Type-C project "LISA-MW" proposal, 10k CPUh at the Italian National HPC center. *2022*
- Travel Grant, Horizon 2020 AHEAD 2020 (High Energy Astrophysics) *2021*
- Travel Grant, American Physical Society, DGRAV Student Travel Grant *2020*
- Travel Grant, Institute of Physics Student Travel fund *2019*
- Travel Grant, Royal Astronomical Society, UK. *2018*

Ph.D. student co-supervisor:

- A. Spadaro, University of Milano-Bicocca. *2022-2025*
- F. Pozzoli, University of Insubria. *2022-2025*

MSc student mentoring:

- M. Piarulli, University of Milano-Bicocca, Master's thesis. *2022-2023*
  (now PhD student at Univ. of Toulouse)
- A. Spadaro, University of Milano-Bicocca, Master's thesis. *2021-2022*
  (now PhD student at Univ. of Milano-Bicocca)
- A. Carzaniga, University of Milano-Bicocca, Master's thesis. *2021-2022*
- A. Geminardi, University of Milano-Bicocca, Master's thesis. *2021-2022*
  (now PhD student at Univ. of Pavia)
- E. Finch, University of Birmingham, Year 4 project. *2018*
- V. Spasova, University of Birmingham, Year 4 project. *2018*

BSc student mentoring:

- H. P. G. Carabajo, University of Milano-Bicocca, Bachelor's thesis. *2023-2024*

Teaching

Taught classes:

- Current and future challenges in GW astronomy, PhD course, Milano-Bicocca (IT). *2023*
- Mathematical physics and gravity (MAF900), Module 3, PhD course, Univ. of Stavanger (NO). *2023*
- Contributed lecture to Astrostatistics (F5802Q014/20), Astrophysics MSc, Milano-Bicocca. *2022*

Tutoring:

- Year 2 Python Computing Lab, Physics BSc, Birmingham. *2017-2021*
- Year 2 Maths for physicists, Physics BSc, Birmingham. *2017-2019*

- Year 4 Physics and communication skills, Physics BSc, Birmingham. *2019*

Academic service

Journal referee

- Physical Review Letters
- The Astrophysical Journal Letters
- Journal of Cosmology and Astroparticle Physics
- Monthly Notices of the Royal Astronomical Society
- Physics Letters D
- The Open Journal of Astrophysics
- NASA Technology Transfer Program

Editorial responsibilities

- MDPI Universe *2024*
  Special Issue *"Challenges and Synergies with Future Gravitational Wave Observatories"*.
- Editorial board Lensing subgroup in the LIGO,Virgo, KAGRA collaboration *2023*
- Co-editor of a Living Review in relativity issue on "LISA data analysis" *2022-2023*

Conference organizer (or committee member)

- IFPU focus week on *"Emerging methods in GW population inference"*, Trieste, IT. *2024*
- *LISA Astrophysics Working Group Conference*, Birmingham, UK. *2022*
- *Gravitational-wave populations: what's next?*, Milan, Italy. *2023*
- *Gravitational-wave Excellence Alliance Training (GrEAT) PhD school*, Birmingham, UK. *2019*
- *Gravitational-wave Open Science Center First Open Data Workshop*, Remote *2019*

Outreach & public engagement

- Development of illustrations and animations for LISA Consortium *2023*
- Development of illustrations and graphic content for LIGO Magazine *2022-2023*
- Development of visualisation interface and skymaps content for GW alerts *2022-2023*
  web and smartphone app. `https://chirp.sr.bham.ac.uk`
- Public engagement talk at the *"Manchester Museum for Science and Industry", UK* *2018*
- Public engagement talks *"A Universe of waves"* and *"Science cafè"*, Italy *2018*
- Organization of biweekly public engagement events *"Astronomy in the city"*, Birmingham, UK *2017-2021*
- Lectures to high-school students, Italy *since 2021*

Professional recognition and service

- Italian Habilitation to Associate Professorship in Astrophysics (ASN 02/C1). *2023*
- Postdoc representative, Department of Physics, University of Milano-Bicocca *2023-2025*
- French Qualification for teaching in Higher Education in Astrophysics (Sec.34). *2023*
  French Ministry of higher education and research (qualification no.23234388826).
- Meetings organizer "PhD meet and greet", University of Birmingham *2021*
- LSC Academic Advisory Committee. *2019-2021*

Memberships

- Italian Center for Supercomputing (ICSC). *since 2021*
- TEONGRAV National Initiative (Gravity Theory) *since 2021*
  Italian National Institute for Nuclear Physics (INFN).
- LIGO, Virgo, Kagra Collaboration, full member. *since 2017*
- LISA Consortium, full member. *since 2018*
- Italian Society of General Relativity and Gravitational Physics (SIGRAV) *since 2021*
- Istituto Nazionale di Astrofisica (INAF) *since 2021*
- American Physical Society (APS), member.
- Italian Physical Society (SIF), member *2021*
- Royal Astronomical Society (RAS), fellow. *2018-2021*

Skills

Programming languages: Python (advanced), Bash (advanced), Go, R, Stan, Julia, Mathematica, C, Qt5.

Other scientific tools: TensorFlow, LIGO lalsuite, LaTeX, git, HPC tools, containerization, continous integration, cloud computing, website development.

Languages: English (fluent), Italian (native), French (basic)

Hobbies

Swimming, running, rock climbing, photography. Sci-fi books, electronic music.

Full publication list

Submitted short-author and collaboration papers which I have substantially contributed to.:

5. *Partial alignment between jets and megamasers: coherent or selective accretion?*.
M. Dotti, **R. Buscicchio**, F. Bollati, R. Decarli, W. Del Pozzo, A. Franchini.
arXiv:2403.18002 [astro-ph.GA].

4. *LISA Definition Study Report*.
M. Colpi, K. Danzmann, M. Hewitson, K. Holley-Bockelmann, et al. (incl. **R. Buscicchio**).
arXiv:2402.07571 [astro-ph.CO].

3. *A weakly-parametric approach to stochastic background inference in LISA*.
F. Pozzoli, **R. Buscicchio**, C. J. Moore, A. Sesana, F. Haardt, A. Sesana.
arXiv:2311.12111 [astro-ph.CO].

2. *The last three years: multiband gravitational-wave observations of stellar-mass binary black holes*.
A. Klein, G. Pratten, **R. Buscicchio**, P. Schmidt, C. J. Moore, E. Finch, A. Bonino, L. M. Thomas, N. Williams, D. Gerosa, S. McGee, M. Nicholl, A. Vecchio.
arXiv:2204.03423 [astro-ph.HE].

1. *Search for gravitational-lensing signatures in the full third observing run of the LIGO-Virgo network*.
LIGO Scientific Collaboration, Virgo Collaboration, KAGRA collaboration.
arXiv:2304.08393 [gr-qc].

Papers in major peer-reviewed journals:

19. *A fast test for the identification and confirmation of massive black hole binary*.
M. Dotti, F. Rigamonti, S. Rinaldi, W. Del Pozzo, R. Decarli, **R. Buscicchio**.
Astronomy & Astrophysics 680 (2023) A69. arXiv:2310.06896 [astro-ph.HE].

18. *Glitch systematics on the observation of massive black-hole binaries with LISA*.
A. Spadaro, **R. Buscicchio**, D. Vetrugno, A. Klein, D. Gerosa, S. Vitale, R. Dolesi, W. J. Weber, M. Colpi.
Physical Review D Phys. Rev. D 108 (2023) 123029. arXiv:2306.03923 [gr-qc].

17. *Implications of pulsar timing array observations for LISA detections of massive black hole binaries*.
N. Steinle, H. Middleton, C. J. Moore, S. Chen, A. Klein, G. Pratten, **R. Buscicchio**, E. Finch, A. Vecchio.
Monthly Notices of the Royal Astronomical Society 525 2 (2023). arXiv:2305.05955 [astro-ph.HE].

16. *Parameter estimation of binary black holes in the endpoint of the up-down instability*.
V. De Renzis, D. Gerosa, M. Mould, **R. Buscicchio**, L. Zanga.
Physical Review D 108 (2023) 024024. arXiv:2304.13063 [gr-qc].

15. *Improved detection statistics for non Gaussian gravitational wave stochastic backgrounds*.
M. Ballelli, **R. Buscicchio**, B. Patricelli, A. Ain, G. Cella.
Physical Review D 107 (2023) 124044. arXiv:2212.10038 [gr-qc].

14. *Detecting non-Gaussian gravitational wave backgrounds: a unified framework*.
R. Buscicchio, A. Ain, M. Ballelli, G. Cella, B. Patricelli.
Physical Review D 107 (2023) 063027. arXiv:2209.01400 [gr-qc].

13. *Detectability of a spatial correlation between stellar-mass black hole mergers and Active Galactic Nuclei in the Local Universe*.
N. Veronesi, E.M. Rossi, S. van Velzen, **R. Buscicchio**.
Monthly Notices of the Royal Astronomical Society 514 2 (2023). arXiv:2203.05907 [astro-ph.HE].

12. *Bayesian parameter estimation of stellar-mass black-hole binaries with LISA*.
R. Buscicchio, A. Klein, E. Roebber, C. J. Moore, D. Gerosa, E. Finch, A. Vecchio.
Physical Review D 104 (2021) 044065. arXiv:2106.05259 [astro-ph.HE].

11. *An Interactive Gravitational-Wave Detector Model for Museums and Fairs*.
S. J. Cooper, A. C. Green, H. R. Middleton, C. P. L. Berry, **R. Buscicchio**, E. Butler, C. J. Collins, C. Gettings, D. Hoyland, A. W. Jones, J. H. Lindon, I. Romero-Shaw, S. P. Stevenson, E. P. Takeva, S. Vinciguerra, A. Vecchio, C. M. Mow-Lowry, A. Freise.
American Journal of Physics 89 (2021) 702–712. arXiv:2004.03052 [physics.ed-ph].

10. *Evidence for hierarchical black hole mergers in the second LIGO–Virgo gravitational-wave catalog*.
C. Kimball, C. Talbot, C.P.L. Berry, M. Zevin, E. Thrane, V. Kalogera, **R. Buscicchio**, M. Carney, T. Dent, H. Middleton, E. Payne, J. Veitch, D. Williams .
Astrophysical Journal Letters 915 (2021) L35. arXiv:2011.05332 [astro-ph.HE].

9. *Testing general relativity with gravitational-wave catalogs: the insidious nature of waveform systematics.*
C. J. Moore, E. Finch, **R. Buscicchio**, D. Gerosa.
iScience 24 (2021) 102577. arXiv:2103.16486 [gr-qc].

8. *LoCuSS: The splashback radius of massive galaxy clusters and its dependence on cluster merger history.*
M. Bianconi, **R. Buscicchio**, G. P. Smith, S. L. McGee, C.P. Haines, A. Finoguenov, A. Babul.
Astrophysical Journal 911 (2021) 136. arXiv:2010.05920 [astro-ph.GA].

7. *Search for Black Hole Merger Families.*
D. Veske, A. G. Sullivan, Z. Marka, I. Bartos, K. R. Corley, J. Samsing, **R. Buscicchio**, S. Marka.
Astrophysical Journal Letters 907 (2021) L48. arXiv:2011.06591 [astro-ph.HE].

6. *Constraining the lensing of binary black holes from their stochastic background.*
R. Buscicchio, C. J. Moore, G. Pratten, P. Schmidt, M. Bianconi, A. Vecchio.
Physical Review Letters 125 (2020) 141102. arXiv:2006.04516 [astro-ph.CO].

5. *Constraining the lensing of binary neutron stars from their stochastic background.*
R. Buscicchio, C. J. Moore, G. Pratten, P. Schmidt, A. Vecchio.
Physical Review D 102 (2020) 081501 . arXiv:2008.12621 [astro-ph.HE].

4. *Measuring precession in asymmetric compact binaries.*
G. Pratten, P. Schmidt, **R. Buscicchio**, L. M. Thomas.
Physical Review Research 2 (2020) 043096. arXiv:2006.16153 [gr-qc].

3. *Populations of double white dwarfs in Milky Way satellites and their detectability with LISA.*
V. Korol, S. Toonen, A. Klein, V. Belokurov, F. Vincenzo, **R. Buscicchio**, D. Gerosa, C. J. Moore, E. Roebber, E. M. Rossi, A. Vecchio.
Astronomy & Astrophysics 638 (2020) A153. arXiv:2002.10462 [astro-ph.GA].

2. *Milky Way satellites shining bright in gravitational waves.*
E. Roebber, **R. Buscicchio**, A. Vecchio, C. J. Moore, A. Klein, V. Korol, S. Toonen, D. Gerosa, J. Goldstein, S. M. Gaebel, T. E. Woods.
Astrophysical Journal Letters 894 (2020) L15. arXiv:2002.10465 [astro-ph.GA].

1. *Label Switching Problem in Bayesian Analysis for Gravitational Wave Astronomy.*
R. Buscicchio, E. Roebber, J. M. Goldstein, C. J. Moore .
Physical Review D 100 (2019) 084041. arXiv:1907.11631 [astro-ph.IM].

Collaboration papers in major peer-reviewed journals, which I have substantially contributed to.:

12. *GWTC-2.1: Deep Extended Catalog of Compact Binary Coalescences Observed by LIGO and Virgo During the First Half of the Third Observing Run.*
LIGO Scientific Collaboration, Virgo Collaboration, KAGRA collaboration.
Physical Review D Phys. Rev. D Physical Review D 109 (2024) 022001. arXiv:2108.01045 [gr-qc].

11. *The population of merging compact binaries inferred using gravitational waves through GWTC-3.*
LIGO Scientific Collaboration, Virgo Collaboration, KAGRA collaboration.
Physical Review X 13 (2021) 011048. arXiv:2111.03634 [astro-ph.HE].

10. *Tests of General Relativity with GWTC-3.*
LIGO Scientific Collaboration, Virgo Collaboration, KAGRA collaboration.
Physical Review D (accepted). arXiv:2112.06861 [gr-qc].

9. *Search for lensing signatures in the gravitational-wave observations from the first half of LIGO-Virgo's third observing run.*
LIGO Scientific Collaboration, Virgo Collaboration, KAGRA collaboration.
Astrophysical Journal Letters (2021) 923. arXiv:2105.06384 [gr-qc].

8. *GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo During the Second Part of the Third Observing Run.*
LIGO Scientific Collaboration, Virgo Collaboration, KAGRA collaboration.
Physical Review X 13 (2023) 041039. arXiv:2111.03606 [gr-qc].

7. *Observation of gravitational waves from two neutron star-black hole coalescences.*
LIGO Scientific Collaboration, Virgo Collaboration.
Astrophysical Journal Letters, 915, L5 (2021). arXiv:2106.15163 [astro-ph.HE].

6. *GWTC-2: Compact Binary Coalescences Observed by LIGO and Virgo During the First Half of the Third Observing Run.*
LIGO Scientific Collaboration, Virgo Collaboration.
Physical Review X 11 (2021) 021053. arXiv:2010.14527 [gr-qc].

5. *Population Properties of Compact Objects from the Second LIGO-Virgo Gravitational-Wave Transient Catalog*.
LIGO Scientific Collaboration, Virgo Collaboration.
Astrophysical Journal Letters 913 (2021) L7. arXiv:2010.14533 [astro-ph.HE].
4. *Upper Limits on the Isotropic Gravitational-Wave Background from Advanced LIGO's and Advanced Virgo's Third Observing Run*.
LIGO Scientific Collaboration, Virgo Collaboration, KAGRA collaboration.
Physical Review D 104 (2021) 022004. arXiv:2101.12130 [gr-qc].
3. *Binary Black Hole Population Properties Inferred from the First and Second Observing Runs of Advanced LIGO and Advanced Virgo* .
LIGO Scientific Collaboration, Virgo Collaboration.
Astrophysical Journal 882 (2019) L24. arXiv:1811.12940 [astro-ph.HE].
2. *Properties and astrophysical implications of the 150 Msun binary black hole merger GW190521*.
LIGO Scientific Collaboration, Virgo Collaboration.
Astrophysical Journal Letters 900 (2020) L13. arXiv:2009.01190 [astro-ph.HE].
1. *GW190521: A Binary Black Hole Merger with a Total Mass of 150* $M_\odot$.
LIGO Scientific Collaboration, Virgo Collaboration.
Physical Review Letters 125 (2020) 101102. arXiv:2009.01075 [gr-qc].

Other publications. These include PhD thesis, and other collaboration papers.:

2. *LISA - Laser Interferometer Space Antenna - Definition Study Report*.
The European Space Agency.
ESA-SCI-DIR-RP-002.
1. *Topics in Bayesian population inference for gravitational wave astronomy*.
R. Buscicchio.
PhD thesis.

Full presentation list

Invited talks marked with *.

Talks at conferences:

17. *From mHz to kHz: stochastic background implications on astrophysical sources and population reconstruction*.
LISA Astrophysics working group workshop, University of Milano-Bicocca, Milano, Italy, Sept 2023.
16. *Non-gaussian gravitational wave backgrounds across the GW spectrum*.
XXV Sigrav conference on general relativity and gravitation, SISSA, Trieste, Italy, Sept 2023.
15.* *LISA SGWB data analysis (session chair)*.
Data Analysis Challenges for SGWB Workshop, CERN, Geneva, Switzerland, Jul 2023.
14.* *Global Fit and foregrounds*.
LISA SGWB detection brainstorming, Univ. of Geneva, Geneva, Switzerland, Jul 2023.
13.* *Beyond functional forms: non-parametric methods. (panelist talk)*.
Gravitational-wave populations: What's next?, Milano-Bicocca, IT, Jul 2023.
12. *The last three years : multiband gravitational-wave observations of stellar-mass binary black holes*.
LISA Astrophysics working group workshop, University of Birmingham, Birmingham, UK, Jun 2022.
11. *The last three years : multiband gravitational-wave observations of stellar-mass binary black holes*.
American Physical Society (APS) April meeting, New York (NY), USA, Apr 2022.
10. *Bayesian parameter estimation of stellar-mass black-hole binaries with LISA*.
XXIV Sigrav conference on general relativity and gravitation, Urbino, Italy, Sept 2021.
9. *Chirp: a web and smartphone application for visualization of gravitational-wave alerts*.
(remote) 14th Amaldi Conference on Gravitational Waves, Jul 2021.
8. *Milky Way Satellites Shining Bright in Gravitational Waves*.
(remote) 13th LISA Symposium, Sept 2020.
7. *Constraining the Lensing of Binary Black Holes from Their Stochastic Background*.
LISA Sprint workshop, Center for Computational Astrophysics, Flatiron Institute, New York (NY), USA, Mar 2020.
6. *Multiple source detection in GW astronomy: the label switching problem*.
30th Texas Symposium, University of Portsmouth, Portsmouth, UK, Dec 2019.
5. *Non-gaussian Stochastic background search with importance sampling*.
LIGO, Virgo, KAGRA March meeting, Lake Geneva Wisconsin, Milwaukee, USA, Mar 2019.

4. *Hierarchical nonparametric density estimation for population inference.*
   LIGO, Virgo, KAGRA September meeting, Warsaw, Poland, Sept 2019.
3. *An improved detector for non-Gaussian stochastic background.*
   Stochastic Background Data Analysis for LISA meeting, Instituto de Fisica Teorica, Madrid, Spain, Jan 2019.
2. *Fast Evaluation of Campbell processes N–point correlation functions.*
   Astro Hack Week: Data Science for Next-Generation Astronomy, Lorentz Center, Leiden, The Netherlands, Aug 2018.
1. *Stochastic Gravitational Wave Background Data Analysis for Radler.*
   5th LISA Cosmology Working Group workshop, Physicum, University of Helsinki, Helsinki, Finland, Jun 2018.

Talks at department seminars:

9.* *GRAF: Gravitational waves data and global fit.*
   Department of Physics, University of Milano-Bicocca, Milan, Italy, Dec 2023.
8.* *LISA global inference: statistical and modelling challenges for the Milky Way.*
   Max Planck Institute for Astrophysics, Munchen, Germany, Nov 2023.
7.* *LISA Global inference: modelling, statistical, and computational challenges.*
   Department of Physics, University of Pisa, Pisa, Italy, Oct 2023.
6.* *Gravitational waves in the many sources, many detectors era.*
   Institute for Mathematics and Physics, University of Stavanger, Stavanger, Norway, Apr 2023.
5.* *The last three years: multiband gravitational-wave observations of stellar-mass binary black holes.*
   Physics Department, Columbia University, New York (NY), USA, 2022.
4.* *Stellar mass binary black holes : what, when, and where.*
   Astroparticule et cosmologie, Universitè Paris Citè, Paris, France, (remote) 2022.
3. *Search for lensing signatures in the gravitational-wave observations from the first half of LIGO-Virgo's third observing run.*
   (remote, on behalf of the LVK collaboration) 2nd European Physical Society Conference on Gravitation, Jul 2021.
2.* *Set the alarm : Bayesian parameter estimation of stellar-mass black-hole binaries with LISA.*
   (remote) Sun Yat-sen University, Zhuhai, China, 2021.
1. *Bayesian parameter estimation of stellar-mass black-hole binaries with LISA.*
   (remote) LISA Data Analysis Conference, 2021.